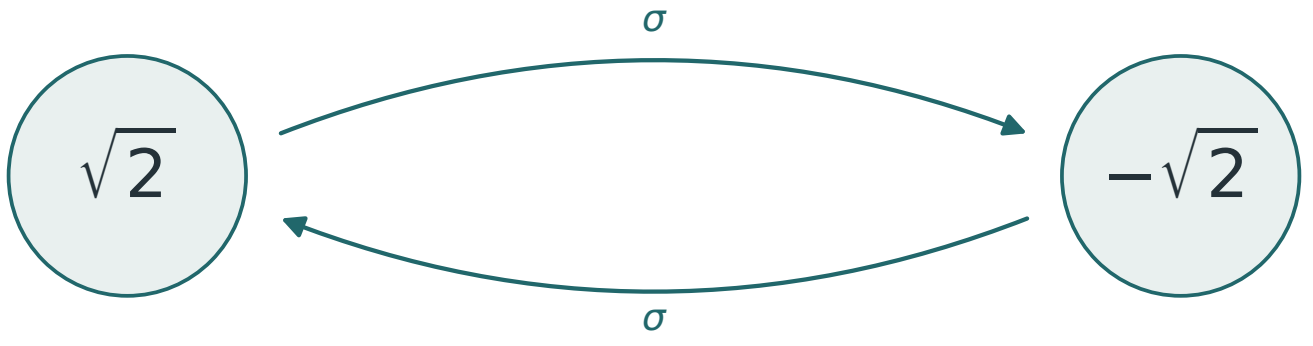
A complete quadratic realization

WRITTEN PROOF / Eisenstein at 2 and two conjugates

$f(x) = x^2 - 2$
$\Delta = 8$

splitting field

$L = \mathbb{Q}(\sqrt{2})$
$[L : \mathbb{Q}] = 2$



$$\sigma^2 = 1, \qquad \mathrm{Gal}(L/\mathbb{Q}) = \{1, \sigma\} \cong C_2$$